INCH-POUND

MIL-R-39007/7F
20 February 1990
SUPERSEDING
MIL-R-39007/7E
15 January 1987

MILITARY SPECIFICATION

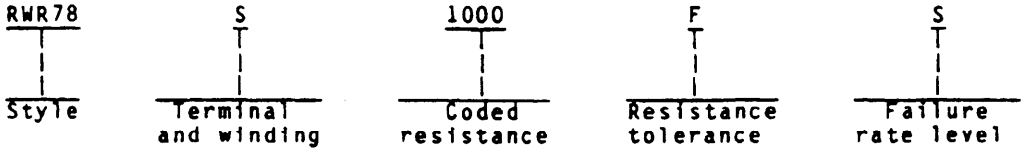
RESISTOR, FIXED, WIRE-WOUND (POWER TYPE), ESTABLISHED RELIABILITY, STYLE RWR78

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RWR78, established reliability, power type, wire-wound, fixed resistor. These resistors are available with terminal types S and W (solderable, inductively wound), and terminal Z (weldable, noninductively wound).

1.2 Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification. Resistors covered by this specification shall be identified by a PIN which shall consist of the style designation, terminal and winding, coded resistive value, tolerance, and failure rate level. The PIN shall be derived in accordance with MIL-R-39007 and shall be in the following form:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-39007 - Resistors, Fixed, Wire-Wound (Power Type), Established Reliability, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39007, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Noninductive windings. Noninductively wound resistors shall be identified by terminal designation N to indicate a solderable terminal and Z to indicate a weldable terminal.

3.3 Minimum resistance value for all terminations (N, Z, S, W). For inductively wound resistors, the minimum resistance value shall be 0.1 ohm. For resistance tolerance B (.1 percent), minimum resistance value shall be .499 ohm.

3.4 Maximum resistance value for S and W (based on use of .001 inch (0.03 mm) nominal diameter wire with a nominal resistivity of 800 ohms per circular mil foot). The maximum resistance value shall be 39,200 ohms. Resistors below 7,000 ohms in value shall be wound with .00175 inch (0.0444 mm) or larger diameter type.

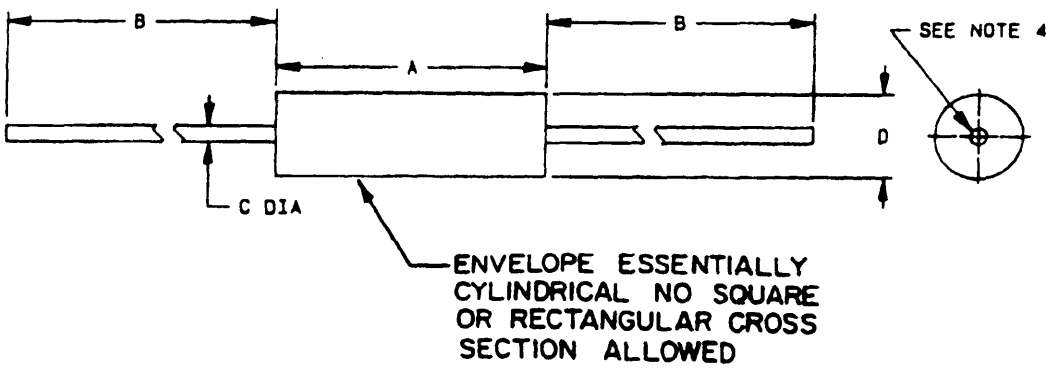
3.5 Maximum resistance value for N and Z type resistors. For resistors having noninductive characteristic N and Z designation, the maximum resistance value shall be 19,600 ohms.

3.6 Power rating. The power rating shall be 10 watts at +25°C.

3.7 Weight. The maximum weight shall be .02646 pound (12 grams); for N type resistors, the maximum weight shall be .02866 pound (13 grams).

3.8 Terminal lead types. This style resistor is available in both solderable (S and N) and weldable (W and Z) leads.

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Inches	mm
.002	0.05
.031	0.79
.032	0.81
.040	1.02
.062	1.57
.375	9.53
1.500	38.10
1.780	45.21

Standard style	Dimensions			
	A ±.062	B, min	C ±.002	D ±.031
RWR78	1.780	1.500	.040	.375

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension A is clean lead to clean lead.
4. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body to be within .032 TIR.

FIGURE 1. Style RWR78 resistor.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39007.

4.2 Short-time overload. The duration of applied voltage shall be 5 seconds. Voltage applied shall be equivalent to 10 times rated wattage.

4.3 Dielectric withstanding voltage.

4.3.1 Atmospheric pressure. The magnitude of test voltage shall be 1,000 volts rms.

4.3.2 Barometric pressure (reduced). The test voltage shall be 300 volts rms.

4.4 Terminal strength. The applied force shall be 10 pounds.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39007.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-39007 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 MIL-R-26 substitution data. Resistors of this specification regardless of their failure rate designation are substitutes for resistors of the same resistance value, tolerance, and appropriate characteristics specified in the inactive specification MIL-R-26/5, style RWR78. Also, resistors of this specification are substitutes for new design for specification MIL-R-26/4, style RW68. However, they are not directly interchangeable.

6.3.1 MIL-R-39007/2 substitution data. For replacement purposes only, use style RWR78 of MIL-R-39007/7 over the deleted RWR68 of MIL-R-39007/2.

6.4 Part or Identifying Number (PIN). The PIN should be structured in accordance with 1.2.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 80, 85
DLA - ES

User activities:
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1183)

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